

Customer Behavior Analysis

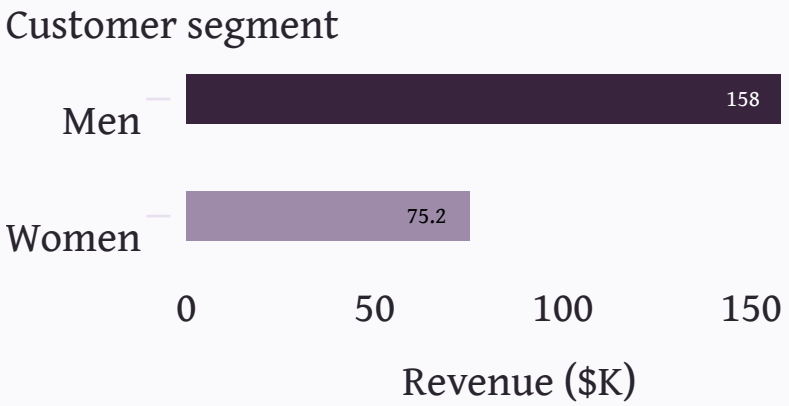
By Mohammed Muqtadir Ahmed

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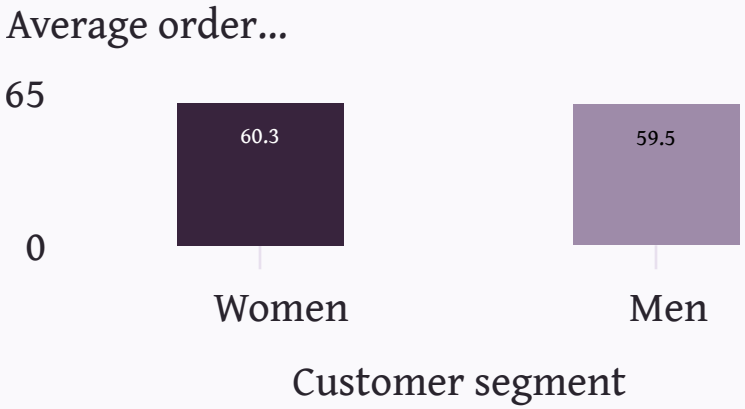
EXECUTIVE READOUT

Revenue is constrained by customer count, not basket size

Average order value stays close to $60 across the major customer and order segments.



3,900 customers generated $233,081 in revenue. Men represent 68% of customers and 68% of revenue, or $157.9K.



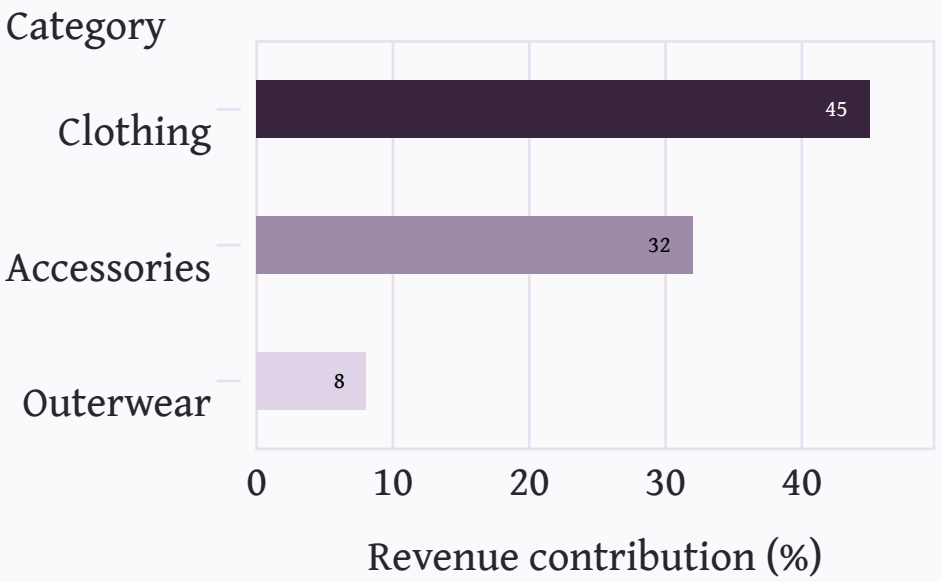
Women generate $75.2K, with nearly identical AOV: $60.25 versus $59.54 for men.

The gap is audience mix, not spending power. Growth should prioritize acquiring more of the right customers and increasing basket size.

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CATEGORY MIX

Clothing and Accessories carry the business



77%

of revenue comes from Clothing and Accessories.

Clothing contributes 45% of revenue, or $104K; Accessories contributes 32%, or $74K.

Outerwear contributes 8% and has the lowest AOV at $57.17.

Category contribution and AOV observations from the dataset.

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RECOMMENDATION 01

Recommendations: make basket size the first growth test

Test the increment before scaling acquisition.

01

Bundle

Test Clothing plus Accessories bundles and a free-shipping threshold around $75.

02

Measure

Measure AOV, units per order and conversion, not revenue alone.

03

Hold out

Use a holdout group so the incremental effect is visible.

NEXT TEST

Target acquisition and redesign subscription. Run a targeted acquisition test for women in Clothing and Accessories.

Replace blanket subscription discounts with perks tied to higher spend: free shipping thresholds, early access or tiered rewards.

Move New customers toward subscription with a post-first-purchase offer.

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RECOMMENDATION 02

Recommendations: protect margin and fix weak lines

Discount discipline

Reduce discount depth on Hat, Sneakers and Coat in a controlled test.

Product review

Review Jeans and Shirt for fit, quality or pricing issues.

Merchandising

Feature Gloves, Sandals and Boots in merchandising.

Outerwear

Review Outerwear pricing, range and premium SKU coverage.

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READ BEFORE ACTING

Important caveats

DATA SCOPE	The dataset has no cost, margin, date or repeat-order data.
DISCOUNTS	Discount findings are observational, not proof of cause.
SEGMENT MIX	Loyal customers are 80% of the dataset because the segment is defined as more than 10 previous purchases.
VALIDATION	Use experiments and tighter segmentation to validate the recommendations.

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APPENDIX

Project Workflow

1

Prepare

Prepare the data in Python: clean fields, fill missing review ratings, create age groups and purchase-frequency days.

2

Load

Load the cleaned DataFrame into SQL Server and verify the table.

3

Answer

Answer 10 business questions with SQL queries covering revenue, discounts, products, shipping, subscription and segments.

4

Visualize

Connect Power BI to SQL Server and build KPI cards, category and age-group visuals, plus slicers for subscription, gender, category and shipment type.

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